

Student Research Questions from Module 0

Project 4 — assigned-question list

These 85 research questions were proposed by students in Module 0. For **Project 4**, choose a number between 1 and 85 before reading through the questions; the question next to that number below is the one you will use. If the wording allows more than two groups or is ambiguous, narrow it to a single two-group (binary) comparison, checking with your instructor if needed.

1. Do students who eat breakfast in the morning have more energy than those who don't?
2. Do *Physarum polycephalum* slime molds prefer oats or barley?
3. Does salt water erode land more effectively than fresh water?
4. Do native plants bring in more pollinators than non native pollinator species?
5. Do full-time undergraduate students at Cal Poly Humboldt who regularly consume energy drinks report higher stress levels than those who do not (measured by a weekly stress questionnaire over a one-semester period)?
6. Do wild deer populations that live near towns have shorter life spans than deer that do not?
7. Do people who take notes digitally or physically perform better on exams?
8. Do people who use more than an hour of social media per day have a shorter attention span than those who use less than an hour a day?
9. Do cat owners spend less money on their pets than dog owners?
10. Do individuals who do not consume any animal products (vegans) have a lower average blood pressure than individuals who consume animal products?
11. Do students who use physical notebooks to take handwritten notes perform better on recall quizzes than students who use laptops to type their notes?
12. Do students who play a musical instrument get better grades than students who don't?
13. Do crows prefer dog/cat food or peanuts? (I feed crows often, and these are both safe foods for them, but I'm curious if there's a preference for one or the other.)
14. Is there a significant difference in the average GPA/academic success of students who identify as morning people vs those who identify as night owls?
15. Do people who study and work at the same time actually manage it well or do they burn out? Is self-care really important or is it an excuse people use to take breaks?
16. Do human populations that consume more sodium than potassium live longer on average, or do human populations that consume more potassium than sodium live longer on average?
17. Do students who interact with the professor get better grades, then those who do not?
18. Do full-time undergraduate students at Cal Poly Humboldt who take handwritten notes get better test scores than those who take typed notes over a period of one semester?
19. Between the Japanese and American Cockroach populations in Sacramento which population is doing the best?
20. What is the diet in box jellyfish vs moon jellyfish?
21. What are the benefits vs problems with modern nuclear energy?
22. Do students listening to jazz perform better on math tests than students listening to pop music?
23. Do parents who spank their children raise more successful children than those that do not?
24. Are people who rent off campus more likely to perceive that they got a better deal for housing than students who live in the dorms?
25. Do students have more energy from drinking coffee instead of tea?
26. Do viruses with longer genomic chains reproduce more efficiently or do viruses with shorter genomic chains reproduce more efficiently?
27. How does postpartum depression differ between low income single mothers and high income partnered mothers?
28. How have salmon populations changed from before Klamath dams fell vs after?
29. Do dogs with merle coloring have more health problems than those without? One could gather data for this by comparing dogs in the same litter with and without merle patterns and their health throughout their lives.
30. Do birds prefer sunflower seeds or dried corn?
31. Do students who attend college full time while working part time jobs have lower gpa's then those who don't work part time jobs?
32. Do people who think cilantro tastes like soap have less taste buds than those who don't? One could gather data by gathering people on both sides and then counting the number of taste buds on each of their tongues.

33. Do students on campus who have been the foster care system find it harder to have self confidence/belonging than those have never been in foster care?
34. Although this is a very complicated type of data to measure, I am interested in how medication hormones affect people assigned female at birth differently. Some people have no reaction while others lose complete control over their thoughts and behavior.
35. Do Sac State students or Cal Poly Humboldt students report more satisfaction in their academic career; is the same true for alumni?
36. Do students who predominantly take online classes have higher GPAs than those taking in person classes?
37. Do students who hand write their notes remember more information than student's who type their notes? This can be measured by comparing scores on a short quiz after a lecture or video.
38. Is jasmine tea more calming than chamomile?
39. Do unemployed students typically get better grades than employed one?
40. What is the overall mental health quality of someone that lives in the city vs more rural places?
41. Does having dishes in the sink raise stress levels?
42. At what rate does short-form content degrade a person's attention span in comparison to watching long-form content?
43. Do students who prioritize regular exercise get higher grades than those who don't?
44. Do open-note exams decrease future recall of the subject, compared to exams based on memorization?
45. Do students who take breaks during study sessions get better grades than those who don't?
46. Do humpback whales sing more in deep or coastal waters?
47. Do students who spend time outside get better grades than those who spend more time inside?
48. Do students on campus who eat breakfast feel better and more energized throughout the day than those who don't?
49. Does spending at least 30 minutes outside improve sleep than those who don't?
50. Do rodent populations enjoy grain, or berries more?
51. Does heat or cold application better combat nausea?
52. Another research question I would be interested in studying is the likeliness of forming a drug addiction versus not forming a drug addiction to nicotine.
53. Do rescued animals that get released back into the wild have the same hunting success as animals born in the wild?
54. How does a student who has one specific location to do work compare to those that do not?
55. Do places in Humboldt county with English Ivy have lower biodiversity than places in Humboldt county without?
56. Do college students who live on campus dorms experience higher stress levels than students living off campus?
57. Another question I'd like to research is how many people feel sympathy and passion for animals versus how many people feel no remorse for murdered animals. (I'll choose between meat farm or roadkill for the topic.)
58. Do students that regularly play games with friends get better grades than those who don't?
59. Do young children who spend more time outdoors less likely to need glasses? One could conduct this study by collecting data from a randomized group of young children on how often they spend time outdoors and then collecting data on how many of them end up needing glasses.
60. Would it be more likely for an arts major to spend more time on coursework compared to a science major?
61. I would like to know the population increase of chinook and coho salmon in the Klamath River after the Klamath River Dam removal.
62. Do runners who train intervals improve their pace faster than those who only train distance?
63. Do students who read for pleasure get higher grades than those who do not?
64. I would like to know do people who wash their dishes right away get better grades?
65. Do cats and dogs not get along or is it something that we just came up with? Do people's pets start acting like their owners or not?
66. One research question on something that I would want an answer on this term would be, do students using their own devices measure better than those that are loaned a device. (Like would student A owning their own laptop make a difference compared to student B, who is loaned a laptop.)
67. Do plant propagations do better in water or in soil?
68. Do animals with smaller hearts live longer on average than those with larger ones?
69. Does rockier soil produce sweeter crops compared to loam soil?

70. Do people who garden have less stress levels than those who do not garden?
71. Are on campus students more likely to arrive to class than one's that have to commute?
72. Does the type of music student hear while study help them or distract them? Does food give people energy or get them sleepy?
73. Do people who eat multiple meals throughout the day have better health than those who eat one big meal a day?
74. What is the success rate of transfer students versus students that complete all 4 years of college at one school?
75. Do people who play video games gain certain cognitive benefits in comparison to people who don't play?
76. Do baseball players who participate in their sport daily have a lower average blood pressure than basketball players who participate in their sport daily?
77. Can people perceive a difference between a 4k and 1440p monitor?
78. How do chlorophyll levels in Humboldt Bay change from winter to summer?
79. At Cal Poly Humboldt, how does the academic performance of full-time undergraduate college students who work part-time jobs (a certain total number of hours) compare to that of full-time college students who do not work, over a period of one semester (measured by end-of-term GPA)?
80. Do Bees produce honey in different quantities with Jazz or Classical Music?
81. How much do Hispanic college students at Cal Poly Humboldt face racism outside of school compared to those who don't?
82. Do traditional artists or digital artists have a higher income?
83. Do biologist students have less animal related phobias than other students?
84. Do people who have pets report higher average moods then people who don't?
85. Do students who participate in physical activity daily perform higher on the AP Statistics exam than those individuals that do not?